Introduction to Machine Learning

What Is Machine Learning?

Machine learning is a subfield of artificial intelligence in which systems learn patterns from data rather than following explicitly programmed rules. Instead of a human writing out every decision, the system is shown many examples and adjusts an internal model so that its predictions improve over time.

A machine learning workflow usually has four stages: collecting and preparing data, choosing a model, training the model on the data, and evaluating how well it generalizes to examples it has never seen. The three broad families of machine learning are supervised learning, unsupervised learning, and reinforcement learning.

Supervised Learning

Supervised learning trains a model on labeled examples, where every input is paired with the correct output. The goal is to learn a mapping from inputs to outputs that generalizes to new, unseen inputs. Because the correct answers are provided during training, the model can measure its error directly and correct itself.

Supervised learning splits into two main task types. Classification predicts a discrete category, such as whether an email is spam or not spam. Regression predicts a continuous value, such as the price of a house. Common supervised algorithms include linear and logistic regression, decision trees, support vector machines, and neural networks.

Unsupervised Learning

Unsupervised learning works with data that has no labels. Rather than predicting a known answer, the model tries to discover structure that is already present in the data. This is useful when labeling is expensive or when the interesting patterns are not known in advance.

Two common unsupervised tasks are clustering and dimensionality reduction. Clustering groups similar examples together; k-means is a widely used clustering algorithm. Dimensionality reduction, such as principal component analysis, compresses data into fewer dimensions while preserving as much information as possible, which helps with visualization and with removing noise.

Reinforcement Learning

Reinforcement learning trains an agent to make a sequence of decisions by interacting with an environment. The agent takes actions, receives rewards or penalties, and adjusts its policy to maximize the total reward it collects over time. Unlike supervised learning, it is not told the correct action; it must discover which actions pay off through trial and error.

Reinforcement learning is a natural fit for problems such as game playing, robotics, and recommendation, where decisions made now affect opportunities later. A central challenge is the trade-off between exploration, trying new actions to gather information, and exploitation, using what is already known to earn reward.

Overfitting and Generalization

A model overfits when it memorizes the training data, including its noise, instead of learning the underlying pattern. An overfit model scores very well on data it has already seen but performs poorly on new data, which defeats the purpose of learning.

To detect overfitting, data is split into a training set and a held-out test set; a large gap between training accuracy and test accuracy is a warning sign. Common remedies include gathering more data, simplifying the model, applying regularization to penalize complexity, and using cross-validation to estimate performance more reliably.